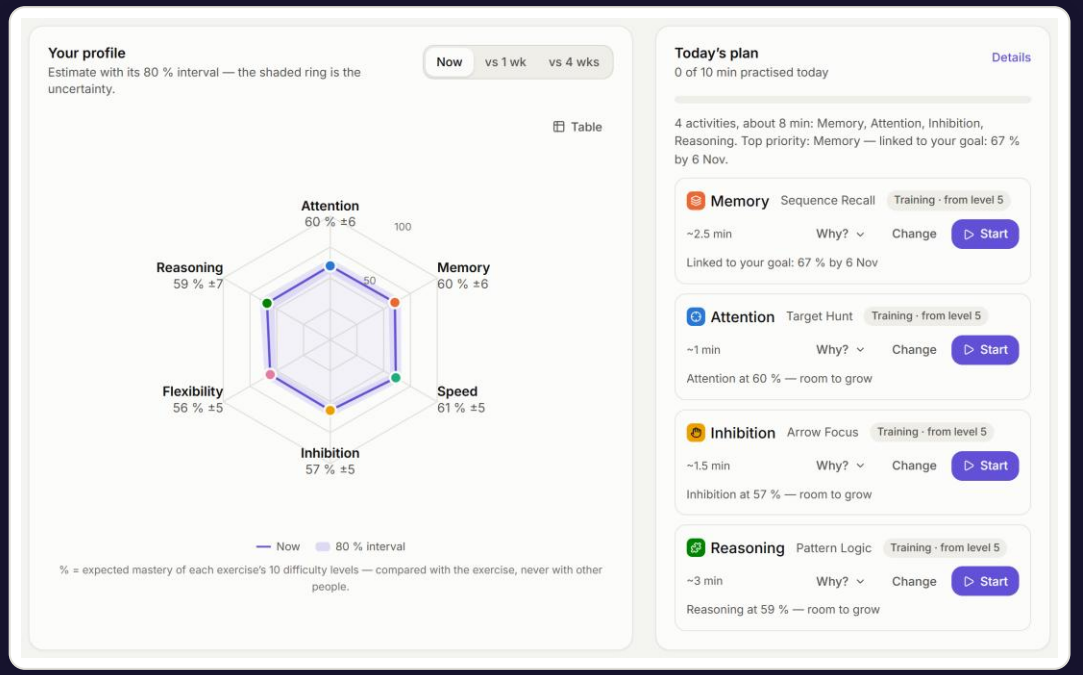
CogniTwin

A cognitive growth agent
with an honest Brain Twin

Assess · Understand · Coach · Improve

Non-clinical · uncertainty-aware · explainable · local-first



Zaineb Ben Fadhl · Rami Bouzid

Team CogniTwin · CISTAC 1.0 — AI4Brain challenge

What is wrong with cognitive apps today

One-time snapshots

A test gives one number, once — but skills change from week to week.

Overclaiming

Precise-looking scores and promises of transfer the evidence does not support.

Black-box AI coaches

Advice that cannot show where it comes from — or is simply invented.

Clinical drift

Scores slide into health language without any validation behind it.

CogniTwin measures continuously, shows uncertainty and exposes evidence. Automated checks help keep coaching within its non-clinical scope.

Our answer: four layers, one rule



Assess

- 6 language-free tasks
- Adaptive check-ins
- Fair device calibration



Understand

- Bayesian % + 80 % interval
- Brain Twin with full history
- Uncertainty grows when idle



Coach

- Evidence-backed insights
- Rules + optional Claude
- Guardrails & “Why?”

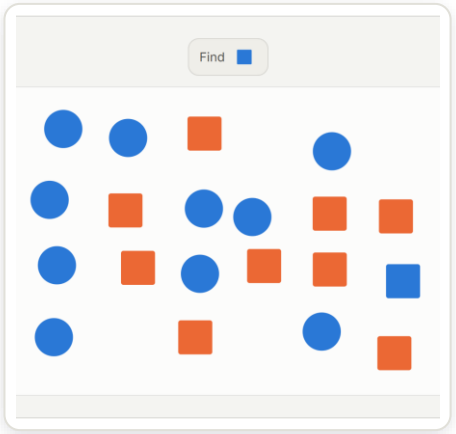


Improve

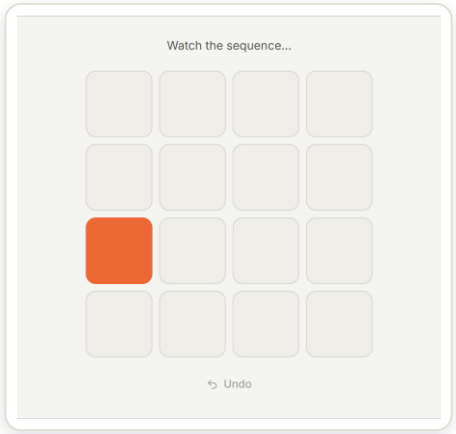
- 80 % target difficulty
- Plan that explains itself
- Goals, missions, projections

One rule: facts come from a tested engine · language comes from AI · uncertainty is shown everywhere

Assess: six adaptive, language-free tasks



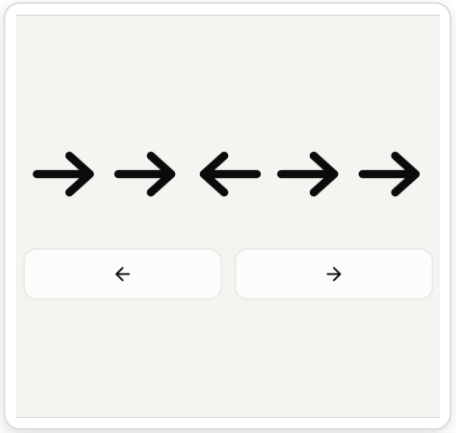
Attention · Target Hunt



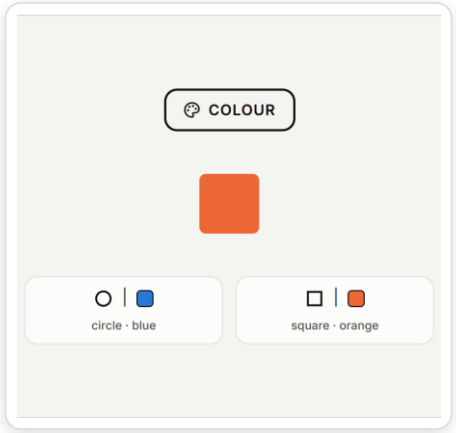
Memory · Sequence Recall



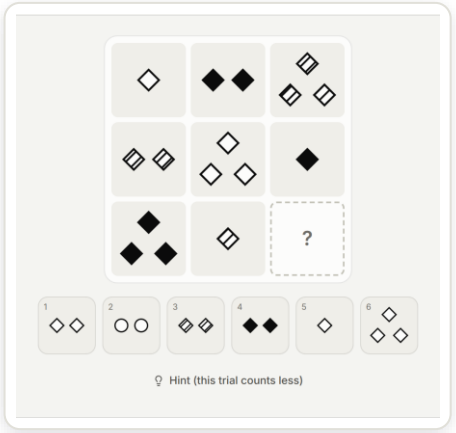
Speed · Symbol Sprint



Inhibition · Arrow Focus



Flexibility · Rule Switch



Reasoning · Pattern Logic

Adaptive check-ins

Each trial picked where it is most informative; stops when precise. ~30 % less error.

Generated, language-free

Fresh stimuli every time: nothing to memorise, no reading or language bias.

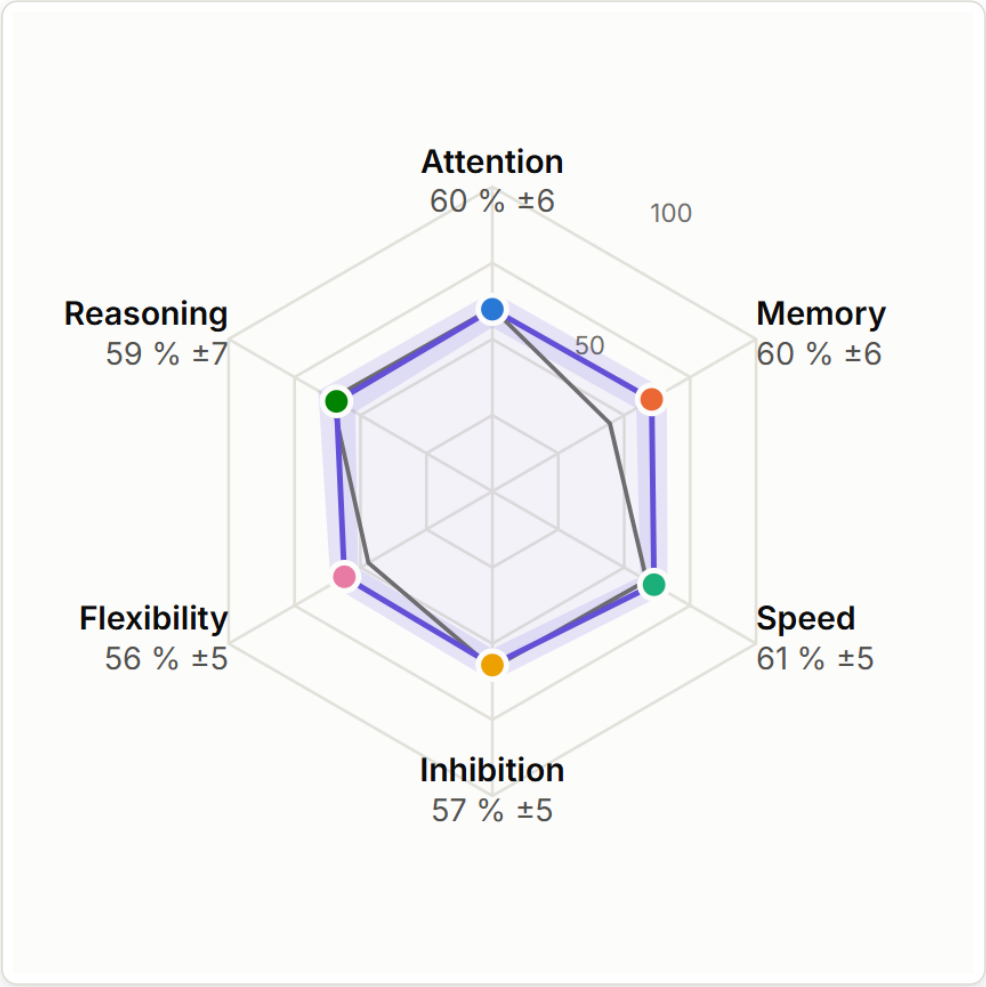
10-tap fairness calibration

Response windows scale to the device and hand speed — phone, trackpad or mouse.

Score, time, errors, behaviour

Every trial: correctness, RT, error type, condition, interruptions, hints.

Understand: a percentage that knows how sure it is



Purple: today with its 80 % interval (shaded) · grey: 4 weeks ago

60 %

80 % interval 55–66 · high confidence

Expected mastery of the task's 10 levels — compared with the task, never with other people.

Bayesian item-response model

Every trial updates an ability estimate and its uncertainty on one scale, whatever the level.

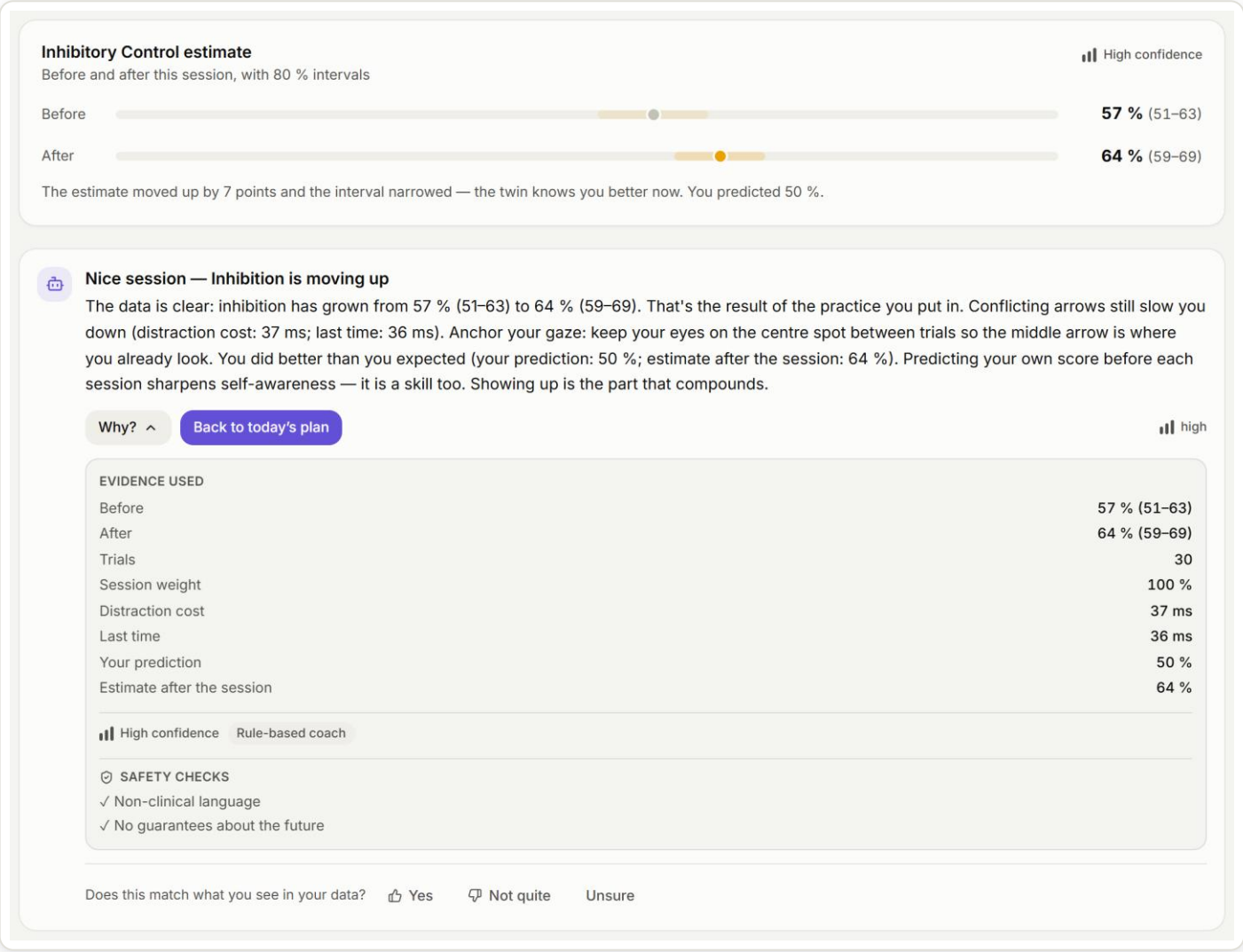
Honest about stale knowledge

The Brain Twin widens the interval when a skill is not practised (73 % vs 33 % coverage).

Full, replayable history

A snapshot after every session: replay, progress curves, heatmaps, session log.

One session, fully explained



1

Before → after

The estimate moves and its 80 % interval narrows: the twin now knows you better.

2

A finding, not a guess

The coach narrates insights the engine computed — with one strategy to try next.

3

“Why?” on every message

Evidence, confidence, source (rules or AI) and the safety checks that passed.

Coach: the engine decides, AI phrases, guardrails verify

Insight engine

Trends, fast errors, fatigue, costs, rule slips — each with evidence + confidence

Narrative

Local rule-based coach or Claude with aggregates and filtered chat

Guardrails

Language checks · cited numbers · confidence cap

“Why?” panel

Evidence, confidence, source and checks on every message

38

term patterns checked in replies and suggestions; semantic accuracy still needs review

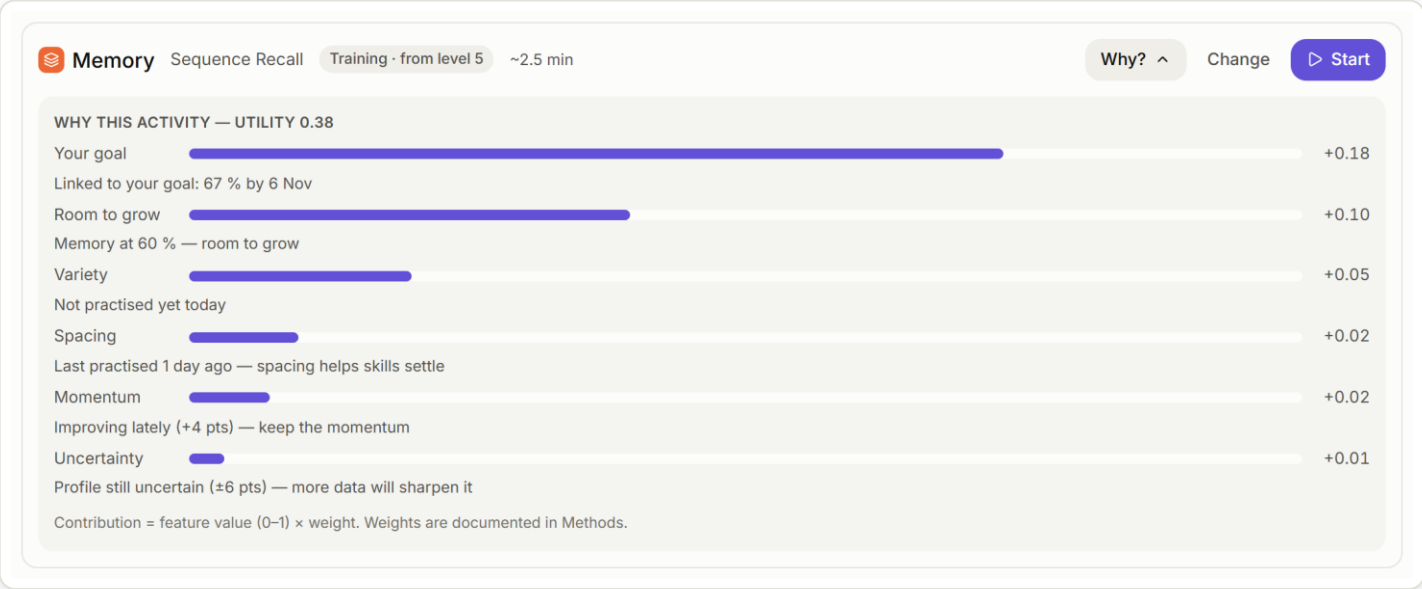
Evidence

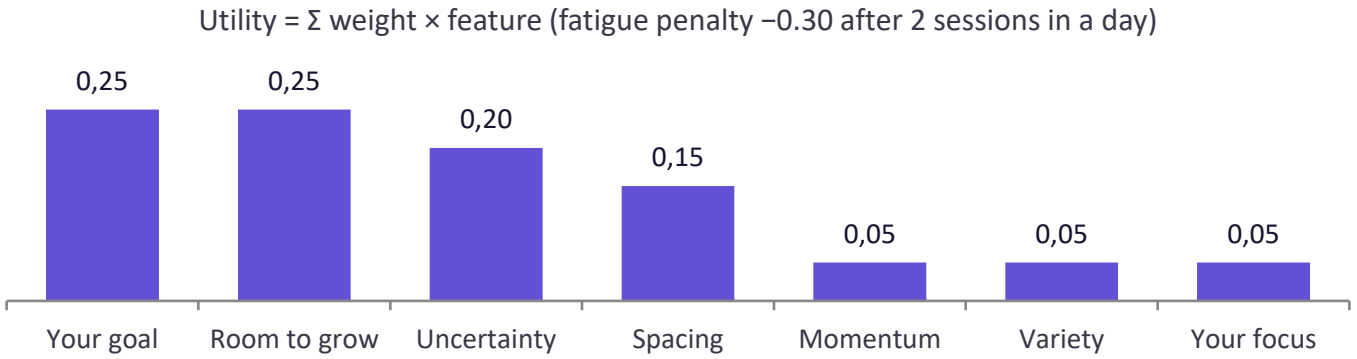
Numbers must match cited source evidence; failed checks trigger regeneration or fallback.

0

health questions sent to the model: a fixed, reviewed reply points to professionals

Improve: a plan that explains itself





≈ 80 % success, every trial

Difficulty follows you; eases after two misses or when strain builds.

An evolving plan

Re-ranked after each session, with a diff: “Reasoning switched to training — profile now confident”.

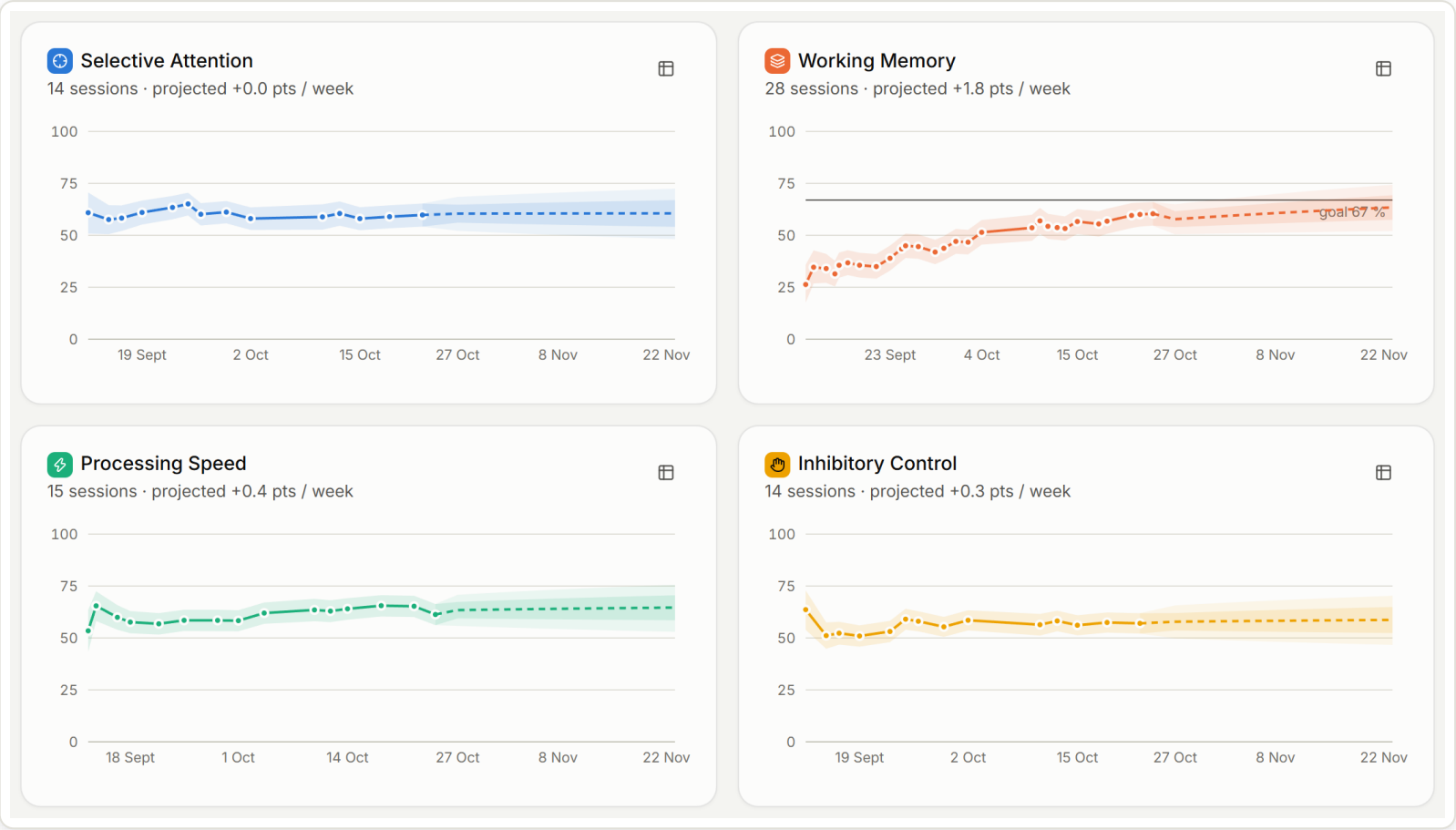
Goals → weekly milestones

Self-set or suggested goals with a path and a model-based chance.

Everyday missions

Real-life practice so gains do not stay inside the app.

Predictive growth — never as certainty



89 %

of true levels 14 days ahead fell inside the 80 % band (simulation, 150 learners)

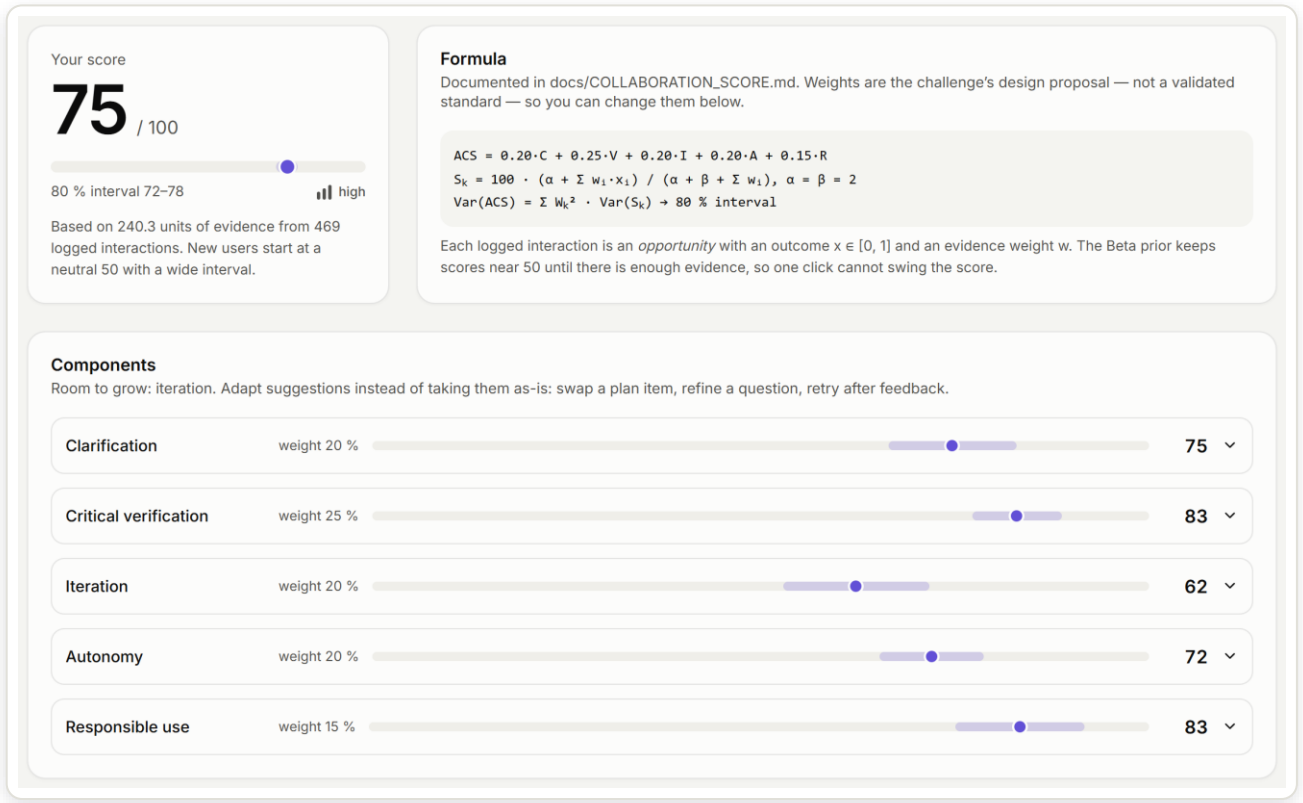
50 / 80 %

fans that widen with time: dashed line = projection, never data

≈ 57 %

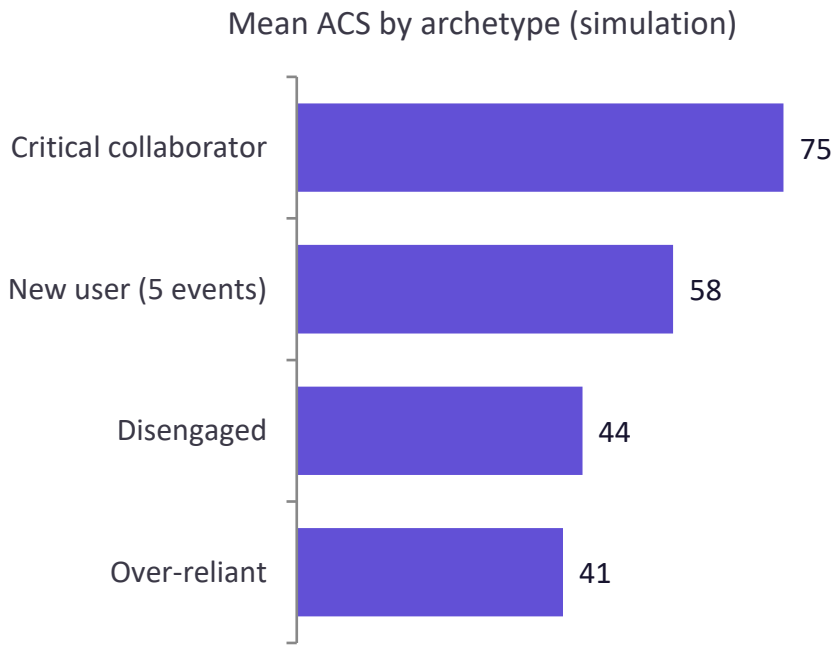
example goal chance — always labelled a model estimate, not a promise

AI Collaboration Score: how well you work with AI



ACS = 0.20 C + 0.25 V + 0.20 I + 0.20 A + 0.15 R

Each component = Beta(2,2)-smoothed share of good outcomes over logged opportunities, with an 80 % interval. Includes an announced "Verify the coach" game.



Emotion-aware and voice coaching — responsibly

Behaviour

always on · weight 1.0

Misses in a row, slowing, rhythm, interruptions

Self-report

optional · weight 0.8

Mood & energy check-in before a session

Camera cues

optional · on-device · weight 0.4

Four averages, no images kept

Working state

in the zone · under strain · getting tired · distracted

Changes only

coach tone · break suggestions · training difficulty — never a score

The camera can never decide alone

A state needs non-camera support (unit-tested). Voice prefers local output; browser services may process text or audio.

Trust by design

Local-first

Stored locally. Optional AI sends aggregates and filtered chat.

Granular consent

AI narrative, voice and camera off by default; every change logged.

Non-biometric by default

Every feature works without camera or microphone.

Retention & deletion

30 / 90 / 365 days or until deleted; delete one session or everything.

Portability

Export all as JSON; import is validated and the twin recomputed.

Non-clinical checks

Coach and UI copy scanned for clinical terms by automated tests.

Validated by simulation with known ground truth

r ≈ 0.9

ability recovered from one 1–3 minute adaptive check-in (6 tasks × 300 users)

–30 %

estimation error with adaptive testing vs random difficulty, same trial count

73 % vs 33 %

interval coverage while learning: Brain Twin vs a model that never forgets

89 %

of true 14-day-ahead levels inside the projected 80 % band

99 % / 0 %

random pressing detected / false alarms on genuine users

127

automated tests pass locally; simulation and build pass

Honest limit: simulation validates the measurement machinery, not construct validity — a pilot study with real participants is the next step.

Built to generalise and to survive

A new dimension = 3 files

Task spec
generator, ladder, metrics

Registry entry
definition, biases, strategies

Task view
rendering + scoring

Picked up automatically

- Brain Twin
- Planner
- Coach & insights
- Projections
- Docs generator
- Validation

Local operation; static hosting
Local coach and storage; serve the static build over HTTP.

Storage & import resilience
IndexedDB → localStorage → memory; imports validated and recomputed.

AI failures degrade gracefully
No key, network error, refusal or unsafe draft → rule-based answer.

Noisy data handled
Interruptions and random pressing are detected and down-weighted.

Thank you

CogniTwin — honest about what it knows, clear about why.

Next steps

- Pilot study and Rasch calibration of item difficulties
- Multilingual coaching (French, Arabic)
- Accessibility study with older adults

Zaineb Ben Fadhl · Rami Bouzid · MIT licence

github.com/zaineb-ben-fadhl/cognitwin youtu.be/fU2qwyaGxMk

